

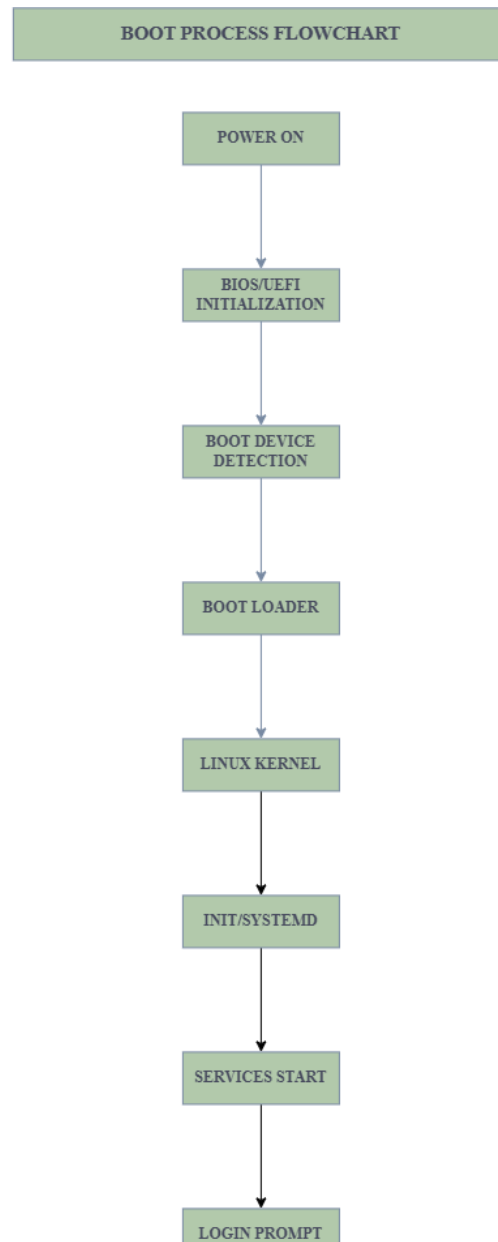


ITEP 414 – SYSTEM ADMINISTRATION AND MAINTENANCE

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Section: BSIT 4E

PART IV. BOOT PROCESS FLOWCHART



The boot process of the Ubuntu system is divided into 8 stages, each stage happens one after the other, starting from when the power is turned on, and ending at the login prompt that is provided for a user when they boot the system:

1. Power On – The system is connected to the power supply, and the CPU fetches instructions from a fixed memory address. Power On – Power is supplied to the system



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and the CPU executes instructions from a fixed memory address thus initiating the boot sequence.

2. BIOS/UEFI Initialization: The firmware (BIOS or UEFI) initializes and performs the Power-On Self-Test (POST) to confirm that hardware components (RAM and CPU, storage controllers) are present, after which control is passed to the operating system.
3. Boot Device Detection – The firmware checks the boot devices (hard disk, USB, network) specified in the priority order and finds a valid boot loader.
4. Boot Loader (GRUB) – GRAND UNIFIED BOOT LOADER (GRUB) is a boot loader that loads from disk. It displays the boot menu (if set up) and loads the Linux kernel and initial RAM disk (initramfs) into memory.
5. Linux Kernel – This initializes and sets up memory allocation, identifies hardware devices, loads drivers for the hardware and mounts the root filesystem.
6. init/systemd – The kernel gives up control to systemd (pid 1), the first user-space process, which takes care of the remainder of the startup sequence.
7. Services Start – systemd starts services and daemons in the proper sequence, depending on their dependencies (networking, SSH, logging, etc.) reaching the desired run level.
8. Login Prompt – After all the necessary services are operating, a login prompt appears, where a user can log in and start a session.

WINDOWS SERVER VS UBUNTU SERVER VS ROCKY LINUX

CRITERIA	WINDOWS SERVER	UBUNTU SERVER	ROCKY LINUX
Licensing	Paid, per-core/CAL licensing	Free, open-source (Canonical offers paid support)	Free, open-source (community-driven, RHEL-compatible)
User Interface	GUI by default (Server Core option for CLI)	CLI by default (GUI optional)	CLI by default (GUI optional)
Package Management	Windows Installer / PowerShell / Server Manager	APT (apt install)	DNF/YUM (dnf install)
Security	Windows Defender, AD-integrated policies, frequent patches	AppArmor, regular security updates, strong community patching	SELinux, enterprise-grade hardening, RHEL security model
Performance	Higher resource overhead due to GUI/services	Lightweight, efficient on low resources	Lightweight, optimized for stability
Enterprise Use Cases	Active Directory, .NET apps, SQL Server, corporate environments	Web hosting, cloud infrastructure, containers, DevOps	Enterprise servers needing RHEL compatibility without licensing cost
Advantages	Deep Microsoft ecosystem integration, familiar GUI	Huge community, frequent updates, great for cloud/Docker	Stable, RHEL binary-compatible,



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			great for orgs replacing CentOS
Disadvantages	Costly licensing, higher resource use	Less enterprise support unless paid	Smaller community than Ubuntu

UEFI is replacing **BIOS**, as it addresses the fundamental problems of **BIOS**. **BIOS** starts in 16-bit mode and only addresses 1MB of memory during start up; **UEFI** is in 32/64-bit mode and accesses memory directly, initializing the system much faster. **BIOS** uses MBR partitioning and supports drives up to 2TB which is a big issue when storage capacity started increasing. GPT is used by **UEFI**, completely eliminating that limit and giving way to many more partitions. Another key reason for security is that **UEFI**'s Secure Boot mechanism ensures that only trusted, signed bootloaders and operating system kernels are loaded, whereas **BIOS** does not have any defense against rootkits and bootkits. **UEFI** also provides a graphical and mouse-driven configuration screen in place of **BIOS**'s text-based menus, as well as capabilities such as network booting and quicker POST (Power-On Self-Test). As a result of these performance, capacity and security benefits, almost all Enterprise servers and modern PCs are now shipped with **UEFI** and **BIOS** mode is reserved for legacy support.